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Lab # 8

Task 1:

Given Grammar:

$S \rightarrow AB$
$A \rightarrow aA \mid \varepsilon$
$B \rightarrow bB \mid \varepsilon$

First ( )	Follow ( )
First (A) = {a, ε}	Follow (A) = {b, $}
First (B) = {b, ε}	Follow (B) = {$}
First (S) = {a, b, ε}	Follow (S) = {$}

Intersection Method:

•) First (aA) ∩ Follow (A)
{a} ∩ {b, $}
= ϕ

•) First (bB) ∩ Follow (B)
{b} ∩ {$}
= ϕ

↳ All intersection = ϕ
" Grammar is LL1 "

Task 2:

Given Grammar:

$E \rightarrow TE'$

$E' \rightarrow +TE' \mid \varepsilon$

$T \rightarrow FT'$

$T' \rightarrow *FT' \mid \varepsilon$

$F \rightarrow id \mid (E)$

Input String: id + id * id $

Start Symbol: E

Initial Stack: $E

Step - by - Step Parsing:

Stack	Input	Action
$E	id+id*id $	$E \rightarrow TE'$
$E'T	id+id*id $	$T \rightarrow FT'$
$E'T'F	id+id*id $	$F \rightarrow id$
$E'T'id	id+id*id $	match id
$E'T'	+id*id $	$T' \rightarrow \varepsilon$
$E'	+id*id $	$E' \rightarrow +TE'$
$E'T+	+id*id $	match +
$E'T	id*id $	$T \rightarrow FT'$
$E'T'F	id*id $	$F \rightarrow id$
$E'T'id	id*id $	match id
$E'T'	*id $	$T' \rightarrow *FT'$
$E'T'F*	*id $	match *
$E'T'F	id $	$F \rightarrow id$
$E'T'id	id $	match id
$E'T'	$	$T' \rightarrow \varepsilon$
$E'	$	$E' \rightarrow \varepsilon$

$	$	Accept

↳ Stack and input both reached "$", which means string can be accepted.

Task 3:

```
#include <iostream>
#include <string>
using namespace std;

string input;
int i = 0;

void S() {
   if(input[i] == 'a') {      // production S → aSb
      cout << "S -> aSb\n";
      i++;                    // match 'a'
      S();                    // recursive call
      if(input[i] == 'b') {  // match 'b'
         i++;
      }
      else {
         cout << "Rejected\n";
         exit(0);
      }
   }
   else {
      cout << "S -> ε\n";    // epsilon production
   }
}

int main() {
   cout << "Input string: ";
   cin >> input;

   S();

   if(i == input.length())
      cout << "String Accepted :)\n";
   else
      cout << "String Rejected :(\n";

   return 0;
}
```

```
C:\Users\HUAWEI\OneDrive\A

Input string: ab
S -> aSb
S -> e
String Accepted :)

--------------------------------
Process exited after 5.213 seconds with return value 0
Press any key to continue . . .
```

```
C:\Users\HUAWEI\OneDrive\A

Input string: aabb
S -> aSb
S -> aSb
S -> e
String Accepted :)

--------------------------------
Process exited after 3.431 seconds with return value 0
Press any key to continue . . .
```

```
C:\Users\HUAWEI\OneDrive\A

Input string: aaabbb
S -> aSb
S -> aSb
S -> aSb
S -> e
String Accepted :)

--------------------------------
Process exited after 26.09 seconds with return value 0
Press any key to continue . . .
```

1- ab (Steps)

1- S()
input [i] = 'a' (use S → aSb)
2- Match 'a', i = 1
Call S()
3- Now input [i] = 'b' → not 'a'
use S → ε
4- Return → now match 'b', i = 2
Parsing complete
i == length (Accepted)

2- aabb

1- S() → aSb
match 'a', i = 1
2- S() → aSb
match 'a', i = 2
3- S() → ε (input [i] = 'b')
4- match 'b', i = 3
match 'b', i = 4 (Accepted)

3- aaabbb

1- S() → aSb
match 'a', i = 1
2- S() → aSb
match 'a', i = 2
3- S() → aSb
match 'a', i = 3
4- S() → ε (input [i] = 'b')
5- match 'b', i = 4
6- match 'b', i = 5
7- match 'b', i = 6 (Accepted)